

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which sum is equal to $\sum_{k=1}^{20} (2k+1)$?

- (A) $1 + 2 + 3 + 4 + \cdots + 20$
- (B) $1 + 3 + 5 + 7 + \cdots + 41$
- (C) $3 + 4 + 5 + 6 + \cdots + 20$
- (D) $3 + 5 + 7 + 9 + \cdots + 41$

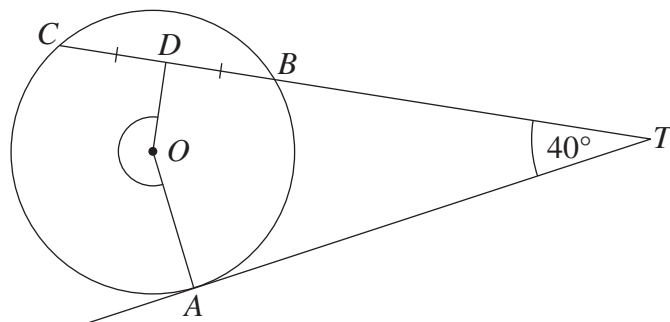
2 What is the remainder when $2x^3 - 10x^2 + 6x + 2$ is divided by $x - 2$?

- (A) -66
- (B) -10
- (C) $-x^3 + 5x^2 - 3x - 1$
- (D) $x^3 - 5x^2 + 3x + 1$

3 Which expression is equivalent to $\frac{\tan 2x - \tan x}{1 + \tan 2x \tan x}$?

- (A) $\tan x$
- (B) $\tan 3x$
- (C) $\frac{\tan 2x - 1}{1 + \tan 2x}$
- (D) $\frac{\tan x}{1 + \tan 2x \tan x}$

- 4 In the diagram, O is the centre of the circle ABC , D is the midpoint of BC , AT is the tangent at A and $\angle ATB = 40^\circ$.



NOT TO
SCALE

What is the size of the reflex angle DOA ?

- (A) 80°
 (B) 140°
 (C) 220°
 (D) 280°
- 5 Which expression is equal to $\int \sin^2 2x \, dx$?

- (A) $\frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$
 (B) $\frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$
 (C) $\frac{\sin^3 2x}{6} + c$
 (D) $\frac{-\cos^3 2x}{6} + c$

6 What is the general solution of the equation $2\sin^2 x - 7\sin x + 3 = 0$?

(A) $n\pi - (-1)^n \frac{\pi}{3}$

(B) $n\pi + (-1)^n \frac{\pi}{3}$

(C) $n\pi - (-1)^n \frac{\pi}{6}$

(D) $n\pi + (-1)^n \frac{\pi}{6}$

7 The displacement x of a particle at time t is given by

$$x = 5\sin 4t + 12\cos 4t.$$

What is the maximum velocity of the particle?

(A) 13

(B) 28

(C) 52

(D) 68

8 A team of 11 students is to be formed from a group of 18 students. Among the 18 students are 3 students who are left-handed.

What is the number of possible teams containing at least 1 student who is left-handed?

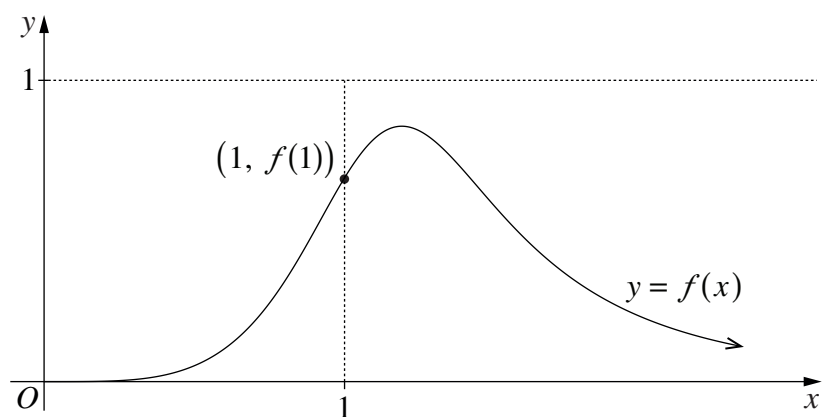
(A) 19 448

(B) 30 459

(C) 31 824

(D) 58 344

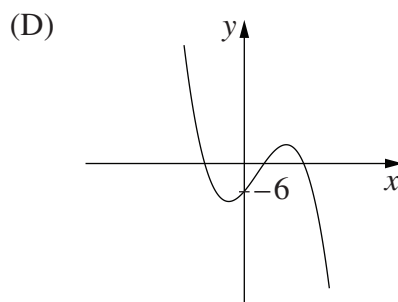
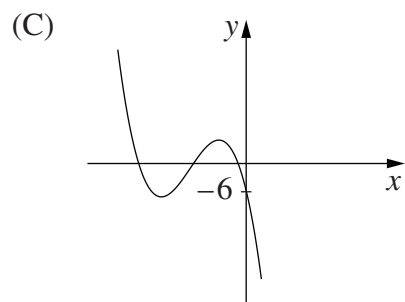
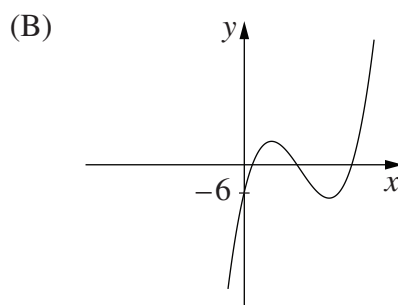
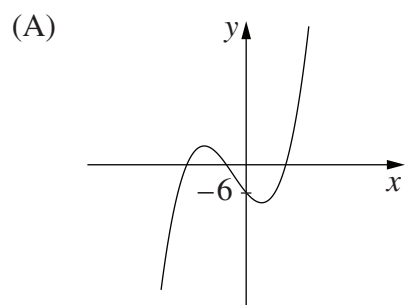
- 9 The diagram shows the graph of $y = f(x)$.



Which of the following is a correct statement?

- (A) $f''(1) < f(1) < 1 < f'(1)$
 (B) $f''(1) < f'(1) < f(1) < 1$
 (C) $f(1) < 1 < f'(1) < f''(1)$
 (D) $f'(1) < f(1) < 1 < f''(1)$
- 10 Consider the polynomial $p(x) = ax^3 + bx^2 + cx - 6$ with a and b positive.

Which graph could represent $p(x)$?



Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

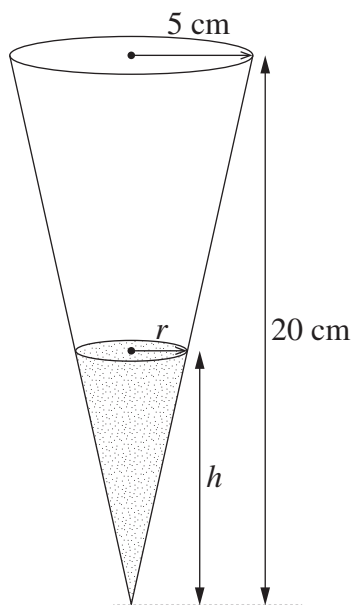
In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the inverse of the function $y = x^3 - 2$. **2**
- (b) Use the substitution $u = x - 4$ to find $\int x\sqrt{x-4} \, dx$. **3**
- (c) Differentiate $3\tan^{-1}(2x)$. **2**
- (d) Evaluate $\lim_{x \rightarrow 0} \left(\frac{2\sin x \cos x}{3x} \right)$. **2**
- (e) Solve $\frac{3}{2x+5} - x > 0$. **3**
- (f) A darts player calculates that when she aims for the bullseye the probability of her hitting the bullseye is $\frac{3}{5}$ with each throw.
- (i) Find the probability that she hits the bullseye with exactly one of her first three throws. **1**
- (ii) Find the probability that she hits the bullseye with at least two of her first six throws. **2**

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows a conical soap dispenser of radius 5 cm and height 20 cm.



At any time t seconds, the top surface of the soap in the container is a circle of radius r cm and its height is h cm.

The volume of the soap is given by $v = \frac{1}{3}\pi r^2 h$.

- (i) Explain why $r = \frac{h}{4}$. 1

- (ii) Show that $\frac{dv}{dh} = \frac{\pi}{16}h^2$. 1

The dispenser has a leak which causes soap to drip from the container. The area of the circle formed by the top surface of the soap is decreasing at a constant rate of $0.04 \text{ cm}^2 \text{ s}^{-1}$.

- (iii) Show that $\frac{dh}{dt} = \frac{-0.32}{\pi h}$. 2

- (iv) What is the rate of change of the volume of the soap, with respect to time, when $h = 10$? 2

Question 12 continues on page 8

Question 12 (continued)

- (b) In a chemical reaction, a compound X is formed from a compound Y . The mass in grams of X and Y are $x(t)$ and $y(t)$ respectively, where t is the time in seconds after the start of the chemical reaction.

Throughout the reaction the sum of the two masses is 500 g.

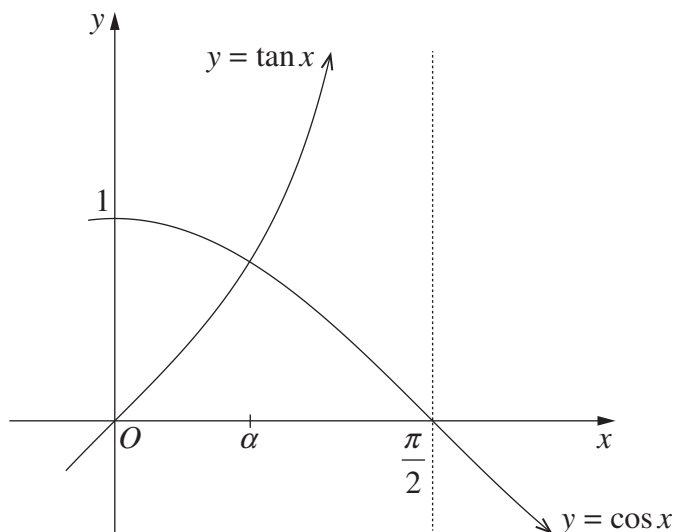
At any time t , the rate at which the mass of compound X is increasing is proportional to the mass of compound Y .

At the start of the chemical reaction, $x = 0$ and $\frac{dx}{dt} = 2$.

(i) Show that $\frac{dx}{dt} = 0.004(500 - x)$. **3**

(ii) Show that $x = 500 - Ae^{-0.004t}$ satisfies the equation in part (i), and find the value of A . **2**

- (c) The graphs of $y = \tan x$ and $y = \cos x$ meet at the point where $x = \alpha$, as shown.



(i) Show that the tangents to the curves at $x = \alpha$ are perpendicular. **2**

(ii) Use one application of Newton's method with $x_1 = 1$ to find an approximate value for α . Give your answer correct to two decimal places. **2**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The tide can be modelled using simple harmonic motion.

At a particular location, the high tide is 9 metres and the low tide is 1 metre.

At this location the tide completes 2 full periods every 25 hours.

Let t be the time in hours after the first high tide today.

- (i) Explain why the tide can be modelled by the function $x = 5 + 4\cos\left(\frac{4\pi}{25}t\right)$. **2**

- (ii) The first high tide tomorrow is at 2 am. **2**

What is the earliest time tomorrow at which the tide is increasing at the fastest rate?

Question 13 continues on page 10

Question 13 (continued)

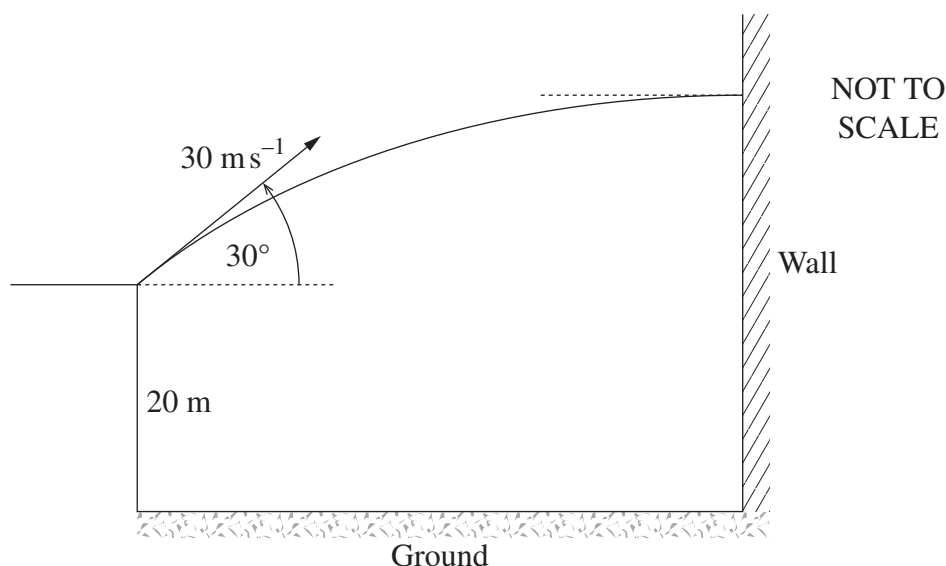
- (b) The trajectory of a projectile fired with speed $u \text{ m s}^{-1}$ at an angle θ to the horizontal is represented by the parametric equations

$$x = ut \cos \theta \quad \text{and} \quad y = ut \sin \theta - 5t^2,$$

where t is the time in seconds.

- (i) Prove that the greatest height reached by the projectile is $\frac{u^2 \sin^2 \theta}{20}$. 2

A ball is thrown from a point 20 m above the horizontal ground. It is thrown with speed 30 m s^{-1} at an angle of 30° to the horizontal. At its highest point the ball hits a wall, as shown in the diagram.



- (ii) Show that the ball hits the wall at a height of $\frac{125}{4}$ m above the ground. 2

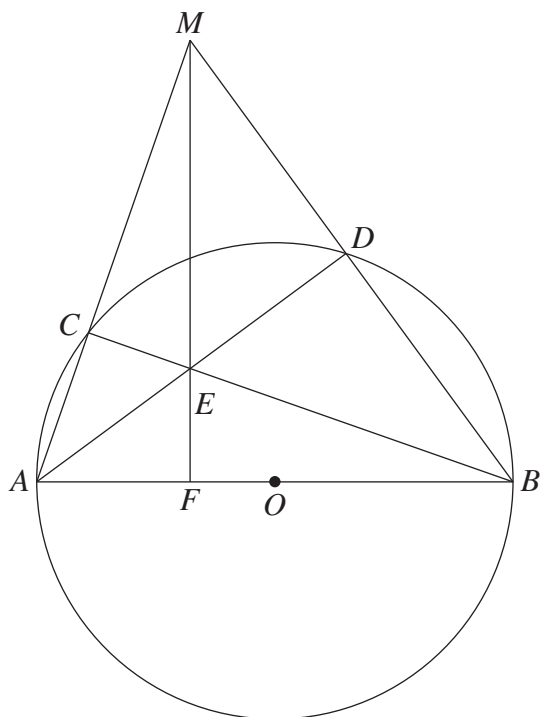
The ball then rebounds horizontally from the wall with speed 10 m s^{-1} . You may assume that the acceleration due to gravity is 10 m s^{-2} .

- (iii) How long does it take the ball to reach the ground after it rebounds from the wall? 2
- (iv) How far from the wall is the ball when it hits the ground? 1

Question 13 continues on page 11

Question 13 (continued)

- (c) The circle centred at O has a diameter AB . From the point M outside the circle the line segments MA and MB are drawn meeting the circle at C and D respectively, as shown in the diagram. The chords AD and BC meet at E . The line segment ME produced meets the diameter AB at F .



Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) Show that $CMDE$ is a cyclic quadrilateral. | 2 |
| (ii) Hence, or otherwise, prove that MF is perpendicular to AB . | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $4n^3 + 18n^2 + 23n + 9$ can be written as **1**

$$(n+1)(4n^2 + 14n + 9).$$

- (ii) Using the result in part (i), or otherwise, prove by mathematical induction that, for $n \geq 1$, **3**

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1).$$

- (b) Consider the expansion of $(1+x)^n$, where n is a positive integer.

- (i) Show that $2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}$. **1**

- (ii) Show that $n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}$. **1**

- (iii) Hence, or otherwise, show that $\sum_{r=1}^n \binom{n}{r}(2r-n) = n$. **2**

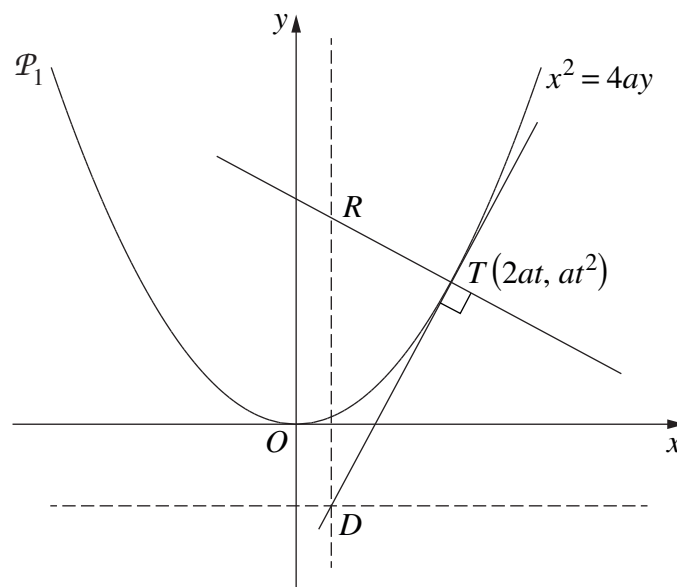
Question 14 continues on page 13

Question 14 (continued)

- (c) The point $T(2at, at^2)$ lies on the parabola $\mathcal{P}_1$ with equation $x^2 = 4ay$.

The tangent to the parabola $\mathcal{P}_1$ at T meets the directrix at D .

The normal to the parabola $\mathcal{P}_1$ at T meets the vertical line through D at the point R , as shown in the diagram.



- | | |
|---|----------|
| (i) Show that the point D has coordinates $\left(at - \frac{a}{t}, -a\right)$. | 1 |
| (ii) Show that the locus of R lies on another parabola $\mathcal{P}_2$. | 3 |
| (iii) State the focal length of the parabola $\mathcal{P}_2$. | 1 |

It can be shown that the minimum distance between R and T occurs when the normal to $\mathcal{P}_1$ at T is also the normal to $\mathcal{P}_2$ at R . (Do NOT prove this.)

- | | |
|--|----------|
| (iv) Find the values of t so that the distance between R and T is a minimum. | 2 |
|--|----------|

End of paper

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REFERENCE SHEET

– Mathematics –

– Mathematics Extension 1 –

– Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

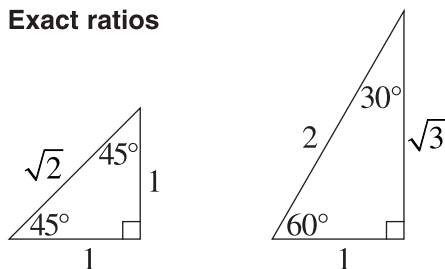
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$\cos(\theta + \phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta\tan\phi}$$

t formulae

If $t = \tan\frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2016 HSC Mathematics Extension 1 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	B
3	A
4	C
5	A
6	D
7	C
8	B
9	A
10	A

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Obtains $x = y^3 - 2$, or equivalent merit	1

Sample answer:

$$y = x^3 - 2$$

Interchange x and y

$$x = y^3 - 2$$

$$\sqrt[3]{x+2} = y$$

$$\therefore f^{-1}(x) = \sqrt[3]{x+2}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains correct integral in terms of u only, or equivalent merit	2
• Attempts to relate du and dx , or equivalent merit	1

Sample answer:

Let $u = x - 4$

$$\frac{du}{dx} = 1 \Rightarrow du = dx$$

$$\begin{aligned}
 \therefore \int x\sqrt{x-4} \, dx &= \int (u+4)u^{\frac{1}{2}} \, du \\
 &= \int \left(u^{\frac{3}{2}} + 4u^{\frac{1}{2}}\right) \, du \\
 &= \frac{2}{5}u^{\frac{5}{2}} + \frac{8}{3}u^{\frac{3}{2}} + C \\
 &= \frac{2}{5}(x-4)^{\frac{5}{2}} + \frac{8}{3}(x-4)^{\frac{3}{2}} + C \\
 &= \frac{2}{5}\sqrt{(x-4)^5} + \frac{8}{3}\sqrt{(x-4)^3} + C
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Obtains $\frac{k}{a^2 + (bx)^2}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{d}{dx}(3 \tan^{-1}(2x)) &= 3 \cdot \frac{2}{1 + (2x)^2} \\ &= \frac{6}{1 + 4x^2}\end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Observes $2 \sin x \cos x = \sin 2x$, or equivalent merit	1

Sample answer:Method 1

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{2 \sin x \cos x}{3x} \right) &= \frac{2}{3} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \cdot \lim_{x \rightarrow 0} \left(\frac{\cos x}{x} \right) \\ &= \frac{2}{\underline{\underline{3}}}\end{aligned}$$

Method 2

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{2 \sin x \cos x}{3x} \right) &= \frac{2}{3} \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \\ &= \frac{2}{\underline{\underline{3}}}\end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Finds the three boundary values for x , or equivalent merit	2
• Attempts to write as a single fraction, or equivalent merit	1

Sample answer:

$$\frac{3}{2x+5} - x > 0$$

$$\frac{3}{2x+5} > x$$

$$(2x+5)^2 \times \frac{3}{\cancel{(2x+5)}} > x(2x+5)^2 \quad \left(x \neq -\frac{5}{2}\right)$$

$$3(2x+5) > x(2x+5)^2$$

$$x(2x+5)^2 - 3(2x+5) < 0$$

$$(2x+5)(2x^2+5x-3) < 0$$

$$(2x+5)(2x-1)(x+3) < 0$$

$$x < -3 \quad \text{or} \quad -\frac{5}{2} < x < \frac{1}{2}$$

Question 11 (e) (continued)

Method 2

$$\frac{3}{2x+5} - x > 0$$

Solve $\frac{3}{2x+5} - x = 0$ for $x \neq -\frac{5}{2}$

$$3 - x(2x+5) = 0$$

$$2x^2 + 5x - 3 = 0$$

$$(2x-1)(x+3) = 0$$

$$x = \frac{1}{2}, \quad x = -3$$



Test	$x = -4$	$x = 0$	$x = 1$
	$\frac{3}{-3} + 4 > 0$	$\frac{3}{5} > 0$	$\frac{3}{7} - 1 < 0$
	✓	✓	✗

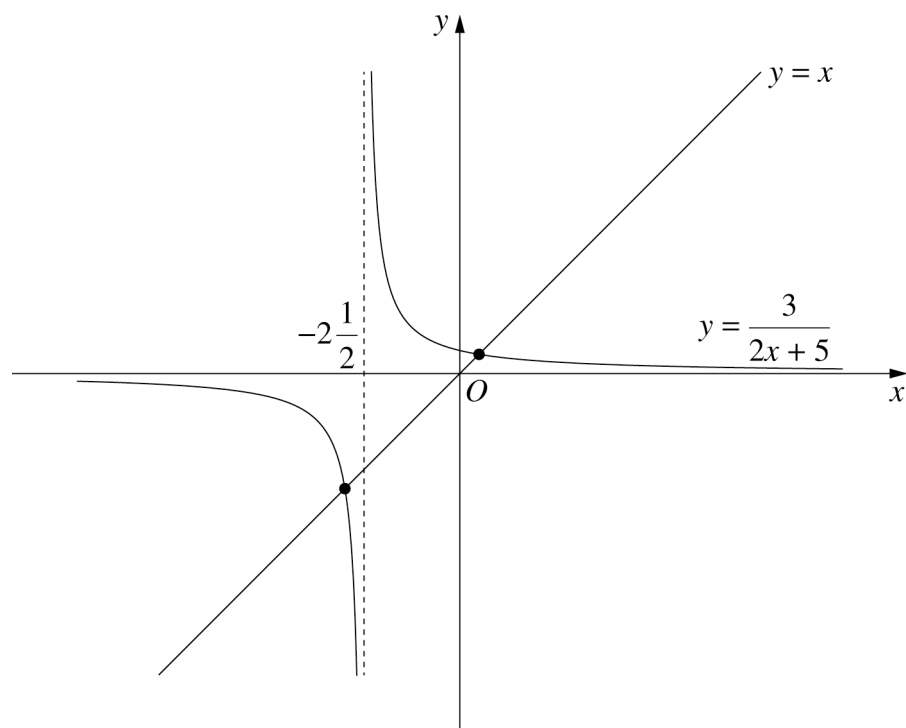
$$x < -3 \quad \text{or} \quad -\frac{5}{2} < x < \frac{1}{2}$$

Question 11 (e) (continued)

Method 3

$$\frac{3}{2x+5} - x > 0$$

$$\frac{3}{2x+5} > x$$



$$\begin{aligned} 3 &= 2x^2 + 5x \\ 2x^2 + 5x - 3 &= 0 \\ (2x - 1)(x + 3) &= 0 \\ x &= \frac{1}{2}, -3 \end{aligned}$$

$$\text{Solution: } x < -3 \text{ or } -2\frac{1}{2} < x < \frac{1}{2}$$

Question 11 (f) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$P(S) = \frac{3}{5}$$

The possibilities are SNN, NNS, NSN

$$\begin{aligned}\therefore P(1S + 2N) &= \left(\frac{3}{5} \times \frac{2}{5} \times \frac{2}{5}\right) \times 3 \\ &= \frac{36}{125}\end{aligned}$$

Question 11 (f) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use complementary events, or equivalent merit	1

*Sample answer:*Method 1

$$\begin{aligned}P(\text{at least 2 bullseyes}) &= {}^6C_2 \left(\frac{3}{5}\right)^2 \left(\frac{2}{5}\right)^4 + {}^6C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^3 + {}^6C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^2 \\ &\quad + {}^6C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^1 + {}^6C_6 \left(\frac{3}{5}\right)^6 \left(\frac{2}{5}\right)^0 \\ &= \frac{432}{3125} + \frac{864}{3125} + \frac{972}{3125} + \frac{2916}{15625} + \frac{729}{15625} \\ &= \frac{2997}{3125}\end{aligned}$$

Method 2

$$\begin{aligned}P(\text{at least 2 bullseyes}) &= 1 - \left[\left(\frac{2}{5}\right)^6 + {}^6C_1 \left(\frac{3}{5}\right) \left(\frac{2}{5}\right)^5 \right] \\ &= \frac{2997}{3125}\end{aligned}$$

Question 12 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

By similar triangles,

$$\frac{r}{5} = \frac{h}{20}$$

$$\therefore r = \frac{h}{\underline{\underline{4}}}$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$v = \frac{1}{3}\pi r^2 h$$

$$v = \frac{1}{3}\pi \left(\frac{h}{4}\right)^2 h$$

$$v = \frac{\pi}{48} h^3$$

$$\therefore \frac{dv}{dh} = \frac{3\pi h^2}{48}$$

$$\Rightarrow \frac{dv}{dh} = \frac{\pi}{\underline{\underline{16}}} h^2$$

Question 12 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use $\frac{dA}{dt}$, or equivalent merit	1

Sample answer:

Show $\frac{dh}{dt} = \frac{-0.32}{\pi h}$

$$A = \pi r^2$$

$$r = \frac{h}{4}$$

$$A = \frac{\pi h^2}{16}$$

$$\frac{dA}{dh} = \frac{\pi h}{8}$$

$$\frac{dA}{dt} = \frac{\pi h}{8} \times \frac{dh}{dt} = -0.04$$

$$\therefore \frac{dh}{dt} = \frac{-0.32}{\pi h}$$

Question 12 (a) (iv)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the chain rule, or equivalent merit	1

Sample answer:

$$\frac{dv}{dh} = \frac{\pi}{16} h^2$$

$$\frac{dh}{dt} = \frac{-0.32}{\pi h}$$

$$\frac{dv}{dt} = \frac{dv}{dh} \cdot \frac{dh}{dt}$$

$$= \frac{\cancel{\pi} h^2}{16} \cdot \frac{-0.32}{\cancel{\pi} h}$$

$$= -0.02h$$

When $h = 10$, $\frac{dv}{dt} = -0.2$

$\therefore$ the volume decreases at a rate of $0.2 \text{ cm}^3 \text{ s}^{-1}$

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Obtains $\frac{dx}{dt} = k(500 - x)$, or equivalent merit	2
• Obtains $\frac{dx}{dt} = ky$ or $y = 500 - x$, or equivalent merit	1

Sample answer:

$$x + y = 500$$

$$y = 500 - x$$

$$\frac{dx}{dt} \propto y$$

$$\therefore \frac{dx}{dt} = ky \quad \text{where } k \text{ is a constant}$$

$$\Rightarrow \frac{dx}{dt} = k(500 - x)$$

When $x = 0$, $\frac{dx}{dt} = 2$

$$\therefore 2 = k(500 - 0)$$

$$\Rightarrow k = \underline{\underline{0.004}}$$

$$\therefore \frac{dx}{dt} = \underline{\underline{0.004(500 - x)}}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Differentiates the given expression, or equivalent merit	1

*Sample answer:*Method 1

$$x = 500 - Ae^{-0.004t}$$

$$\frac{dx}{dt} = A \times 0.004e^{-0.004t}$$

$$= 0.004(Ae^{-0.004t})$$

$$= 0.004(500 - x) \quad \text{as required.}$$

Now, $t = 0$, $\frac{dx}{dt} = 2$, $x = 0$

$$\therefore 0 = 500 - Ae^{-0.004(0)}$$

$$A = 500$$

Method 2

$$\frac{dx}{dt} = 0.004(500 - x)$$

$$\int \frac{dx}{500 - x} = \int 0.004 dt$$

$$-\ln(500 - x) = 0.004t + C$$

$$500 - x = Ae^{-0.004t} \quad \text{where } A \text{ is a constant.}$$

$$\therefore x = \underline{\underline{500 - Ae^{-0.004t}}}$$

When $t = 0$, $x = 0$, $\Rightarrow A = \underline{\underline{500}}$

$$\therefore x = \underline{\underline{500(1 - e^{-0.004t})}}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Relates the slopes of both curves at α , or equivalent merit	1

Sample answer:

$$y = \tan x$$

$$\therefore \frac{dy}{dx} = \sec^2 x$$

$$\text{At } x = \alpha, \quad \frac{dy}{dx} = \sec^2 \alpha$$

$$y = \cos x$$

$$\therefore \frac{dy}{dx} = -\sin x$$

$$\text{At } x = \alpha, \quad \frac{dy}{dx} = -\sin \alpha$$

$$\text{Now product of gradients} = \sec^2 \alpha \times (-\sin \alpha)$$

$$= -\frac{\sin \alpha}{\cos \alpha} \cdot \frac{1}{\cos \alpha}$$

$$= -\frac{\tan \alpha}{\cos \alpha}$$

$$= -1 \quad \text{since } \tan \alpha = \cos \alpha \text{ at the point of intersection}$$

$\therefore$ Tangents to the curves are perpendicular at $x = \alpha$.

Question 12 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds a relevant function to use Newton's method, or equivalent merit	1

Sample answer:Let $f(x) = \tan x - \cos x$

$$f'(x) = \sec^2 x + \sin x$$

Using Newton's method to solve $\tan \alpha = \cos \alpha$, with $x_0 = 1$.

$$\begin{aligned}
 x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= 1 - \frac{f(1)}{f'(1)} \\
 &= 1 - \frac{\tan 1 - \cos 1}{\sec^2 1 + \sin 1} \\
 &\approx 1 - \frac{1.0171054}{4.2669898} \\
 &\approx 1 - 0.238366 \\
 &= \underline{0.76} \quad (2 \text{ decimal places})
 \end{aligned}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct explanation	2
• Explains why the amplitude is 4, or equivalent merit	1

Sample answer:

From reference sheet

$$x = b + a \cos(nt + \alpha)$$

High tide = maximum = 9

Low tide = minimum = 1

$$\therefore a = \frac{9-1}{2} = 4$$

$$\text{Period, } T = \frac{2\pi}{n}; \quad \therefore \frac{25}{2} = \frac{2\pi}{n}$$

$$\Rightarrow n = \frac{4\pi}{25}$$

$t = 0$ at the high tide so $b = 5$ and $\alpha = 0$

$$\therefore x = 5 + 4 \cos\left(\frac{4\pi t}{25}\right)$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to find t when $\ddot{x} = 0$, or equivalent merit	1

Sample answer:

$$x = 5 + 4 \cos\left(\frac{4\pi t}{25}\right)$$

$$\frac{dx}{dt} = -4 \cdot \frac{4\pi}{25} \sin\left(\frac{4\pi t}{25}\right)$$

$$\frac{d^2x}{dt^2} = \frac{-16\pi}{25} \cdot \frac{4\pi}{25} \cos\left(\frac{4\pi t}{25}\right)$$

Fastest when $\frac{d^2x}{dt^2} = 0$

Increasing fastest when

$$\cos\left(\frac{4\pi t}{25}\right) = 0$$

$$\therefore \frac{4\pi t}{25} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$t = \frac{25}{8}, \frac{75}{8}, \dots$$

Tide is going *out* when $t = \frac{25}{8}$

$$\therefore \text{Increasing when } t = \underline{\underline{\frac{75}{8}}} = 9\frac{3}{8}.$$

So $9\frac{3}{8}$ hours after 2 am makes the time $11:22\frac{1}{2}$ am (11:22:30 am)

(or $11\frac{3}{8}$ hours after midnight).

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Finds $\dot{y}$, or equivalent merit	1

Sample answer:

$$x = ut \cos \theta \quad y = ut \sin \theta - 5t^2$$

$$\dot{x} = u \cos \theta \quad \dot{y} = u \sin \theta - 10t$$

Maximum height if $\dot{y} = 0 \Rightarrow u \sin \theta - 10t = 0$

$$u \sin \theta = 10t$$

$$t = \frac{u \sin \theta}{10}$$

Substitute into $y = ut \sin \theta - 5t^2$

$$\begin{aligned} \therefore y &= u \sin \theta \cdot \frac{u \sin \theta}{10} - 5 \left(\frac{u \sin \theta}{10} \right)^2 \\ &= \frac{u^2 \sin^2 \theta}{10} - \frac{u^2 \sin^2 \theta}{20} \end{aligned}$$

$$\therefore \text{Maximum height} = \frac{u^2 \sin^2 \theta}{20}$$

Question 13 (b) (ii)

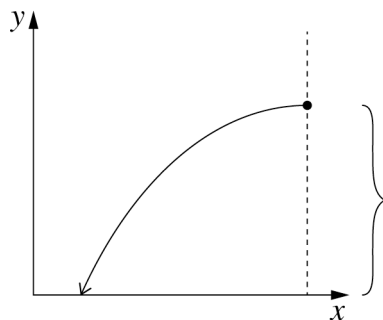
Criteria	Marks
• Provides correct solution	2
• Shows that the maximum height above the projection point is $\frac{45}{4}$, or equivalent merit	1

Sample answer:

$$\begin{aligned} \text{Maximum height} &= 20 + \frac{30^2 \sin^2 30}{20} \\ &= 20 + \frac{900 \times \frac{1}{4}}{20} \\ &= \frac{125}{4} \end{aligned}$$

Question 13 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains correct equation for y in terms of t , or equivalent merit	1

Sample answer:

Taking left ($\leftarrow$) to be positive and origin as start point.

$$\dot{x} = 10 \quad \ddot{y} = -10$$

$$x = 10t \quad \dot{y} = -10t \quad (\text{since } \dot{y}(0) = 0)$$

$$(\text{since } x(0) = 0) \quad y = -5t^2 \quad (\text{since } y(0) = 0)$$

$$y = -31.25 \Rightarrow -31.25 = -5t^2$$

$$t^2 = 6.25$$

$$t = \pm 2.5, \quad t > 0$$

$$t = 2.5$$

$\therefore$ Ball hits the ground after 2.5 seconds.

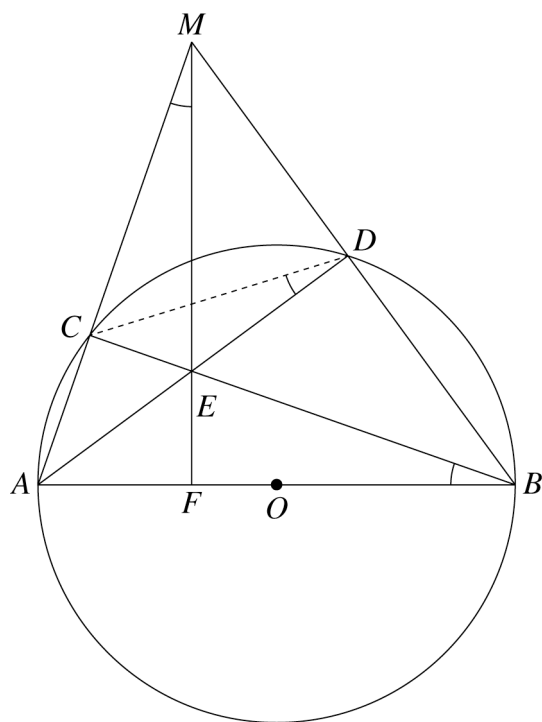
Question 13 (b) (iv)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$t = 2.5 \Rightarrow x = 10(2.5) \\ = 25$$

$\therefore$ The ball lands 25 m from the base of the wall.

Question 13 (c)**Question 13 (c) (i)**

Criteria	Marks
• Provides correct solution	2
• Shows that one of the angles at C or D is a right angle, or equivalent merit	1

Sample answer:

$$\angle ADB = 90^\circ, \quad \angle BCA = 90^\circ \quad (\angle \text{ in a semicircle})$$

$$\angle MDE = 90^\circ, \quad (\angle \text{ sum, straight line})$$

$\therefore CMDE$ is a cyclic quadrilateral (Exterior $\angle ACB$ equal to opposite interior $\angle MDE$)

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows $\angle ABC = \angle ADC$, or equivalent merit	1

Sample answer:Method 1Construct CD Let $\angle ABC = x^\circ$

$$\therefore \angle ADC = x^\circ \quad (\angle\text{s in the same segment})$$

Similarly $\angle CME = x^\circ$

$$\angle CAB = 90^\circ - x^\circ \quad (\angle \text{sum } \triangle ABC)$$

$$\begin{aligned} \therefore \angle MFA &= 180^\circ - \angle CME - \angle CAB && (\angle \text{sum } \triangle MFA) \\ &= 180^\circ - (x^\circ) - (90^\circ - x^\circ) \\ &= 90^\circ \end{aligned}$$

$$\therefore MF \perp AB$$

Method 2Construct CD as shown in the diagram

$$\angle ABC = \angle ADC \quad (\angle\text{s in the same segment})$$

$$\angle ADC = \angle EMC \quad (\angle\text{s in the same segment as } CDME \text{ are concyclic points})$$

$$\triangle CEM \parallel \triangle FEB \quad (\text{equiangular})$$

$$\therefore \angle MCE = \angle BFE = 90^\circ \quad (\text{corresponding } \angle\text{s})$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}\text{R.T.P. } 4n^3 + 18n^2 + 23n + 9 \\ = (n + 1)(4n^2 + 14n + 9)\end{aligned}$$

$$\begin{aligned}\text{RHS} &= n(4n^2 + 14n + 9) + 1(4n^2 + 14n + 9) \\ &= 4n^3 + 14n^2 + 9n + 4n^2 + 14n + 9 \\ &= 4n^3 + 18n^2 + 23n + 9 \\ &= \text{LHS}\end{aligned}$$

OR

$$\begin{aligned}\text{LHS} &= 4n^3 + 4n^2 + 14n^2 + 14n + 9n + 9 \\ &= 4n^2(n + 1) + 14n(n + 1) + 9(n + 1) \\ &= (n + 1)(4n^2 + 14n + 9) \\ &= \text{RHS}\end{aligned}$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct proof	3
• Establishes induction step, or equivalent merit	2
• Establishes initial case, or equivalent merit	1

Sample answer:

$$\text{Let } T_n = (2n-1)(2n+1), S_n = \frac{1}{3}n(4n^2 + 6n - 1)$$

$$\text{R.T.P. } 1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1)$$

$$\begin{aligned} \text{When } n = 1, \quad \text{LHS} &= 1 \times 3 & \text{RHS} &= \frac{1}{3}(1)(4(1)^2 + 6(1) - 1) \\ &= 3 & &= \frac{1}{3} \times 9 \\ & & &= 3 \end{aligned}$$

 $\therefore$ True for $n = 1$

$$\text{Assume true for some integer } k, \text{ ie } S_k = \frac{1}{3}k(4k^2 + 6k - 1)$$

and attempt to show it is also true for $k + 1$

$$\begin{aligned} \text{ie, try to show } S_{k+1} &= \frac{1}{3}(k+1)(4(k+1)^2 + 6(k+1) - 1) \\ &= \frac{1}{3}(k+1)(4k^2 + 8k + 4 + 6k + 6 - 1) \\ &= \frac{1}{3}(k+1)(4k^2 + 14k + 9) \end{aligned}$$

$$\text{Now } S_{k+1} = S_k + T_{k+1}$$

$$\begin{aligned} &= \frac{1}{3}k(4k^2 + 6k - 1) + (2(k+1)-1)(2(k+1)+1) \\ &= \frac{1}{3}k(4k^2 + 6k - 1) + (2k+1)(2k+3) \\ &= \frac{1}{3}(4k^3 + 6k^2 - k) + 4k^2 + 8k + 3 \\ &= \frac{1}{3}(4k^3 + 6k^2 - k + 12k^2 + 24k + 9) \\ &= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9) \\ &= \frac{1}{3}(k+1)(4k^2 + 14k + 9) \quad (\text{From part (i)}) \end{aligned}$$

as required

 $\therefore$ Hence the result is true for all positive integers n , by induction.

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$$

Let $x = 1$ then

$$(1+1)^n = \binom{n}{0} + \binom{n}{1}(1) + \binom{n}{2}(1)^2 + \cdots + \binom{n}{n}(1)^n$$

$$\therefore 2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:Differentiate $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$, with respect to x .

$$n(1+x)^{n-1} = \binom{n}{1} + \binom{n}{2}(2x) + \binom{n}{3}(3x^2) + \cdots + \binom{n}{n}nx^{n-1}$$

and substitute $n = 1$ into this equation

$$n(1+1)^{n-1} = \binom{n}{1} + \binom{n}{2}(2(1)) + \binom{n}{3}(3(1)^2) + \cdots + \binom{n}{n}(n(1)^{n-1})$$

$$\therefore n \cdot 2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Attempts to write the right-hand side in terms of the previous two parts, or equivalent merit	1

Sample answer:Method 1

$$\begin{aligned}
\sum_{r=1}^n \binom{n}{r} (2r - n) &= \binom{n}{1}(2 - n) + \binom{n}{2}(4 - n) + \binom{n}{3}(6 - n) + \cdots + \binom{n}{n}(2n - n) \\
&= 2\binom{n}{1} + 4\binom{n}{2} + 6\binom{n}{3} + \cdots + 2n\binom{n}{n} - n\left[\binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}\right] \\
&= 2\left\{\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + n\binom{n}{n} - n\left[\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}\right] - \binom{n}{0}\right\} \\
&= 2(n \cdot 2^{n-1}) - n(2^n - 1) \\
&= n2^n - n2^n + n \\
&= n
\end{aligned}$$

Method 2

$$\begin{aligned}
&\sum_{r=1}^n \binom{n}{r} (2r - n) \\
&= 2\sum_{r=1}^n \binom{n}{r} r - n\sum_{r=1}^n \binom{n}{r} \\
&= 2\sum_{r=0}^n \binom{n}{r} r - n\left(\sum_{r=0}^n \binom{n}{r} - 1\right) \\
&= 2 \cdot n \cdot 2^{n-1} - n(2^n - 1) \\
&= n
\end{aligned}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

From reference sheet, equation of tangent is $y = tx - at^2$

At D on the directrix, $y = -a$

$$\Rightarrow -a = tx - at^2$$

$$tx = at^2 - a$$

$$x = at - \frac{a}{t}$$

$$\therefore D \text{ is } \left(at - \frac{a}{t}, -a \right)$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Attempts to eliminate the parameter to find the Cartesian equation of $\mathcal{P}_2$, or equivalent merit	2
• Finds coordinates of R , or equivalent merit	1

Sample answer:

From reference sheet, equation of normal is $x + ty = at^3 + 2at$

At R (above D), $x = at - \frac{a}{t}$

$$\Rightarrow at - \frac{a}{t} + ty = at^3 + 2at$$

$$ty = at^3 + at + \frac{a}{t}$$

$$y = at^2 + a + \frac{a}{t^2}$$

$$= a\left(t^2 + 1 + \frac{1}{t^2}\right)$$

$$\therefore R \text{ is } \left(a\left(t - \frac{1}{t}\right), a\left(t^2 + 1 + \frac{1}{t^2}\right)\right)$$

Method 1

$$x = a\left(t - \frac{1}{t}\right)$$

$$x^2 = a^2\left(t^2 - 2 + \frac{1}{t^2}\right)$$

$$\frac{x^2}{a^2} = t^2 + \frac{1}{t^2} - 2$$

$$\frac{x^2}{a^2} + 2 = t^2 + \frac{1}{t^2}$$

$$y = a\left(\left(t^2 + \frac{1}{t^2}\right) + 1\right)$$

$$= a\left(\frac{x^2}{a^2} + 2 + 1\right)$$

$$y = \frac{x^2}{a} + 3a$$

$$ay = x^2 + 3a^2$$

Method 2

$$\frac{x}{a} = t - \frac{1}{t}$$

$$y = a\left(t^2 + 1 + \frac{1}{t^2}\right)$$

$$= a\left(\left(t - \frac{1}{t}\right)^2 + 3\right)$$

$$= a\left(\frac{x^2}{a^2} + 3\right) = \frac{x^2}{a} + 3a$$

$$\therefore (y - 3a)a = x^2$$

Locus is $x^2 = a(y - 3a)$.

A parabola! Vertex $(0, 3a)$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

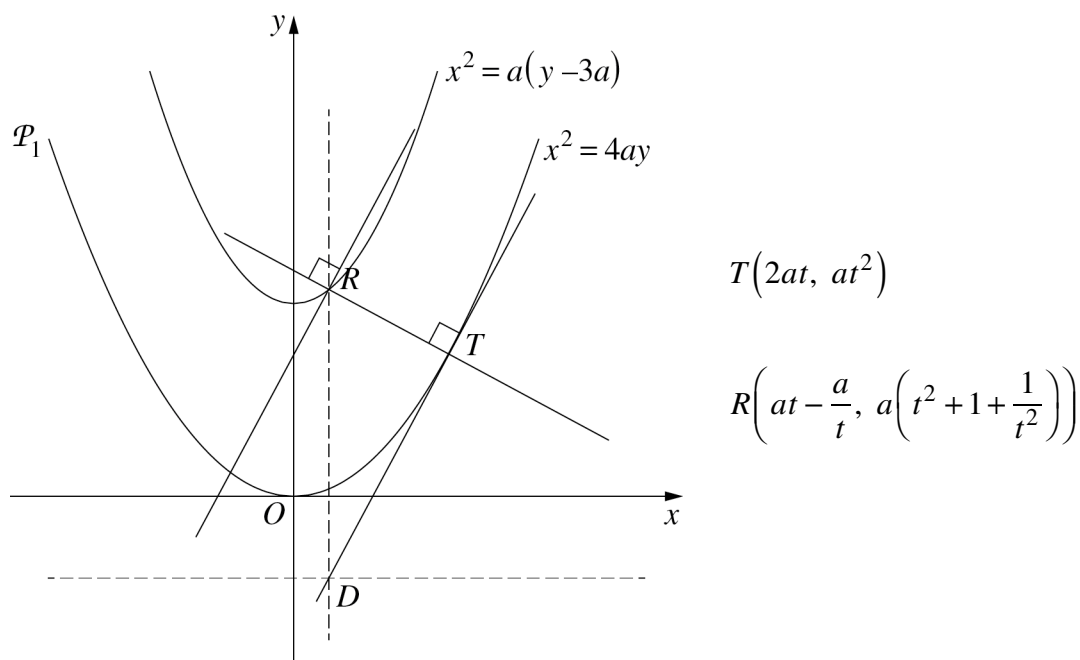
$$x^2 = a(y - 3a)$$

$$\text{ie } x^2 = 4\left(\frac{a}{4}\right)(y - 3a)$$

$$\therefore \text{Focal length } \frac{a}{4}$$

Question 14 (c) (iv)

Criteria	Marks
• Provides correct solution	2
• Finds slope of $\mathcal{P}_2$ at R , or equivalent merit	1

Sample answer:

A common normal will occur when the tangents at T and R are parallel.

Now gradient at T is t

$$\text{For } R, \quad \frac{x^2}{a} = y - 3a$$

Question 14 (c) (iv) continued

$$y = \frac{x^2}{a} + 3a$$

$$y' = \frac{2x}{a}$$

$$\text{So } y' = \frac{2\left(at - \frac{a}{t}\right)}{a}, \text{ since } x = at - \frac{a}{t},$$

$$= 2\left(t - \frac{1}{t}\right)$$

$$\text{So gradient at } R \text{ is } 2t - \frac{2}{t}$$

$$\text{For tangents to be parallel, } t = 2t - \frac{2}{t}$$

$$t^2 = 2t^2 - 2$$

$$2 = t^2$$

$$t = \pm\sqrt{2}$$

2016 HSC Mathematics Extension 1

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	7.1	H5, HE7
2	1	16.2E	PE3
3	1	5.7E	P3, PE6
4	1	2.10E, 2.9E	PE3
5	1	13.6E	HE6
6	1	15.2E, 5.9E	HE4
7	1	14.3, 14.4E	HE5
8	1	18.1E	PE3
9	1	10.1, 10.4	H6
10	1	16.1 E, 16.3E	PE3

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	15.1E	HE4
11 (b)	3	11.5E	HE6
11 (c)	2	15.5E	HE4
11 (d)	2	5.7E, 13.4E	HE7
11 (e)	3	1.4E	PE3
11 (f) (i)	1	3.2, 18.2E	HE3
11 (f) (ii)	2	18.2E	HE3
12 (a) (i)	1	2.3	P2, PE6
12 (a) (ii)	1	14.1E	HE5
12 (a) (iii)	2	14.1E	HE5
12 (a) (iv)	2	14.1E	HE5
12 (b) (i)	3	14.2E	HE3
12 (b) (ii)	2	14.2E	HE3
12 (c) (i)	2	6.2, 10.7, 13.5	H5, HE4
12 (c) (ii)	2	16.4E	HE7
13 (a) (i)	2	14.4E	HE3
13 (a) (ii)	2	14.4E	HE3

Question	Marks	Content	Syllabus outcomes
13 (b) (i)	2	14.3E	HE3
13 (b) (ii)	2	14.3E	HE3
13 (b) (iii)	2	14.3E	HE3
13 (b) (iv)	1	14.3E	HE3
13 (c) (i)	2	2.10E, 2.7E	PE3
13 (c) (ii)	2	2.10E, 2.8E	PE3
14 (a) (i)	1	1.3	P4, PE6
14 (a) (ii)	3	7.4E	HE2
14 (b) (i)	1	17.1E, 17.3E	HE7
14 (b) (ii)	1	8.7, 17.1E, 17.3E	HE4, HE7
14 (b) (iii)	2	17.1E, 17.3E	HE7
14 (c) (i)	1	9.6E	PE3
14 (c) (ii)	3	9.6E	PE3
14 (c) (iii)	1	9.6E	PE3
14 (c) (iv)	2	10.7, 9.6E	HE7